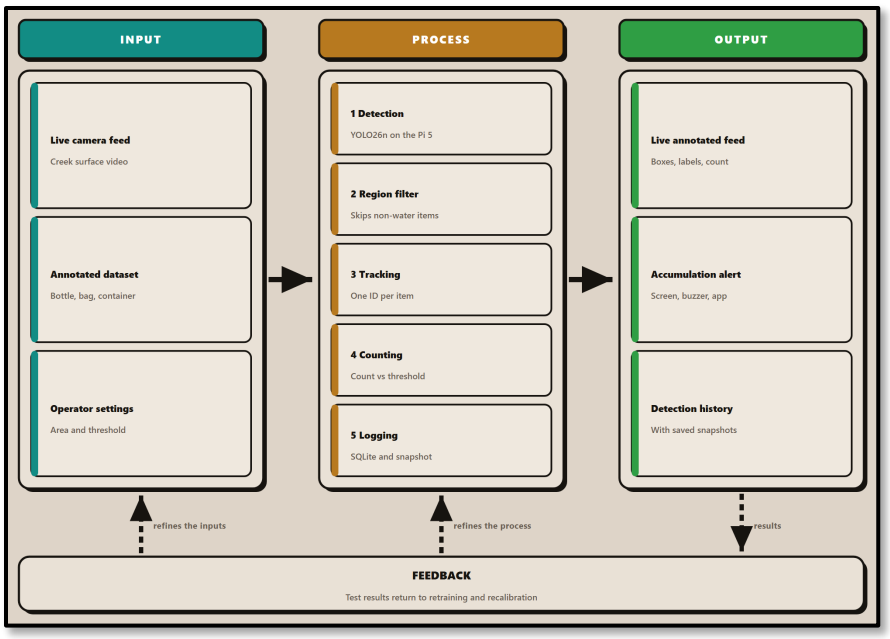
II. LITERATURE REVIEW

2.1 Conceptual Framework of the Study

Figure 1

Conceptual Framework of Garbage-Guard



The conceptual framework of Garbage-Guard is illustrated in Figure 1, and is structured in an IPO or Input-Process-Output model with a feedback path. The process column is expanded to show all of the system block, because these are what make this system a detector, not just a frame labelling system.

There are three inputs to the system. The first is the live camera feed from the network camera mounted above the creek surface that provides frames on an ongoing basis, not on a schedule. The second is the annotated data set which is divided into three classes (bottles, bags and food containers). The third one is a set of operator settings: the region of interest drawn on the water, the number of items to be accumulated (default is twenty), and the alert preferences.

The processing is done in four stages on a Raspberry Pi 5. Detection is done by a YOLO26n convolutional network that is exported to NCNN format that the board runs

fastest. Tracking will allow and maintain one identity per object, frame to frame – they will no longer flitter in and out of view, but have a solid box with a label. Any detection that is not in the traced water area is then discarded by the region filtering, and so are not included in the count, which helps to prevent the inclusion of fixed objects at the edge of the bank. Counting involves taking the number of accumulated objects present in the area of each frame, smoothing it as the median of the last 15 frames (whereby it is not moved by a single object that is missed or falsely detected), and comparing it against the threshold. Logging writes each alert, together with its snapshot, to a local database and synchronizes it to the cloud.

The point to emphasize is the region filter and the smoothed count, this is what takes this detector over to a monitoring system. If a detector detects twenty bounding boxes in one frame, it doesn't mean that twenty objects are there, because one frame may have a false detection or may not detect an object, and a system that can't exclude the bank will report a concrete wall as garbage while it is in frame.

The system has three outputs. The live annotated feed with the bounding boxes, class labels and running count are displayed on the touchscreen dashboard. An accumulation alert is triggered on screen, on the buzzer and sent to the Android app when the threshold is exceeded. All recorded events are kept in a detection history that can be searched, with a snapshot saved for each, this is how a reading can be checked afterwards.

Evaluation is based on standards of ISO/IEC 25010:2023: functional suitability, performance efficiency, and usability, for evaluating the input and process phases. A weak per-class results lead to further annotation and retraining, while latency measured during operation dictates the frame sampling rate, and comments from the operator on the interface are used to guide design of the dashboard. The feedback path is illustrated, since this is what occurred in this study: The detection model is trained using multiple versions and the accumulation threshold is not predetermined but is tuned on site.

2.2 Review of Related Literature and Studies

Sx

A. Environmental Garbage Issues in Philippine Waters

Based on the UNEP (2023) or United Nations Environment Programme, World Bank (2022), and Acot et al. (2022), urban waterways such as rivers and creeks play a critical role in the lives of people living in urban centers.

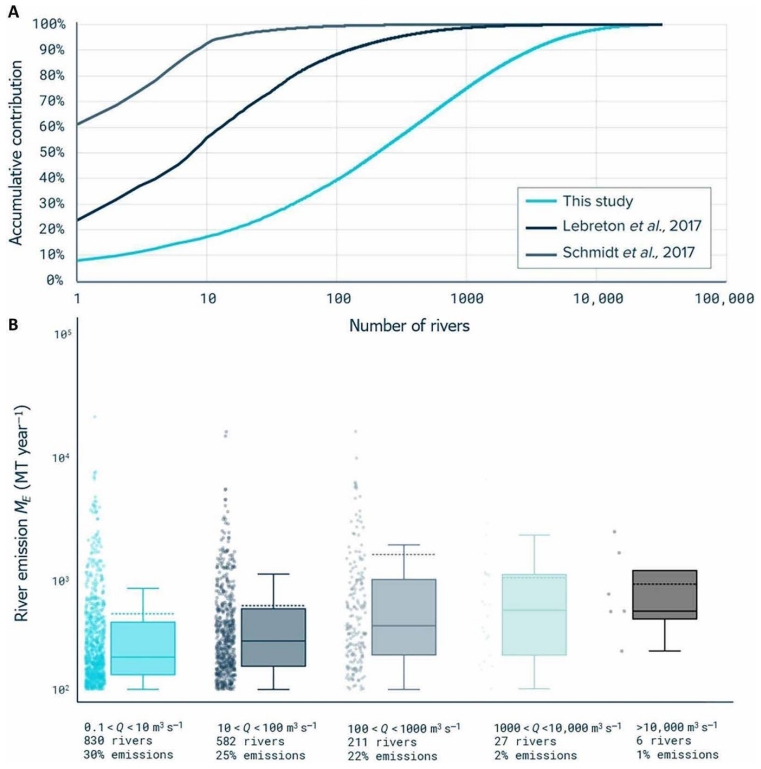
However, due to the effects of rapid urbanization and an ever-increasing population, the amount of solid garbage generated has increased over the years, particularly in the Philippines.

The researchers emphasized that a huge volume of garbage generated in the urban area ends up in the waterway due to inappropriate garbage disposal mechanisms. The researchers' study results revealed that waterways are considered accumulation points for solid garbage originating from the surrounding communities.

However, the researchers' study focused on the garbage management policies and environmental effects rather than the technological aspect. These researchers' studies are related to the present research because they emphasized the need to monitor garbage pollution in the waterway, such as a creek.

Figure 2

Cumulative Contribution of Rivers to Global Plastic Emission



Note. From "More than 1000 rivers account for 80% of global riverine plastic emissions into the ocean," by L. J. J. Meijer, T. van Emmerik, R. van der Ent, C. Schmidt and L. Lebreton, 2021, Science Advances, 7(18). CC BY-NC.

Rivers as the Pathway for Ocean Plastic (Meijer et al., 2021)

According to Meijer et al. (2021), rivers are the pathways that check the transportation of garbage from land to ocean. The research showed global modeling and data finding of environmental data to see sources and pathways of garbage in oceans. The research findings saw that a serious amount of garbage was entering the ocean coming from rivers that are mainly in heavy populated environments. However the research was heavily leaning on data analysis and modeling techniques, which is not effective in checking groups of garbage in creeks. This research is similar to the current study, because it shows the dangers of rivers in the transportation of garbage materials.

Mismanaged Plastic Waste and Urban Flooding (MacAfee & Löhr, 2023)

MacAfee and Löhr (2023) checked the interaction of non-organized plastic waste (MPW) and flooding in rapid urbanization and climate change. The review indicates that the rise in the amount of plastic waste in drain channels increases the risks of flooding, especially in coastal cities MacAfee and Löhr (2023). The limitation of the research is based on the fact that it is based on existing literature without offering technological solutions. The research is relevant in the sense that it emphasizes the need for waste monitoring in the prevention of environmental and social hazards.

Identifying Plastic Waste in Marine Ecosystems (Das et al., 2025)

According to Das et al. (2025), the paper "Plastic Waste in Marine Ecosystems: Identification Techniques and Policy Interventions" reviewed a lot of research papers on marine plastic waste, focusing on techniques and policies that can be intervened. The findings showed that polyethylene and polypropylene are main plastic waste, and the best way to identify plastic waste is by using infrared spectroscopy and AI recognition. The limitations of this research is that it only depended on literature studies and not experiments. The research paper is still relevant to the current study because it showed that waste in water can still be managed.

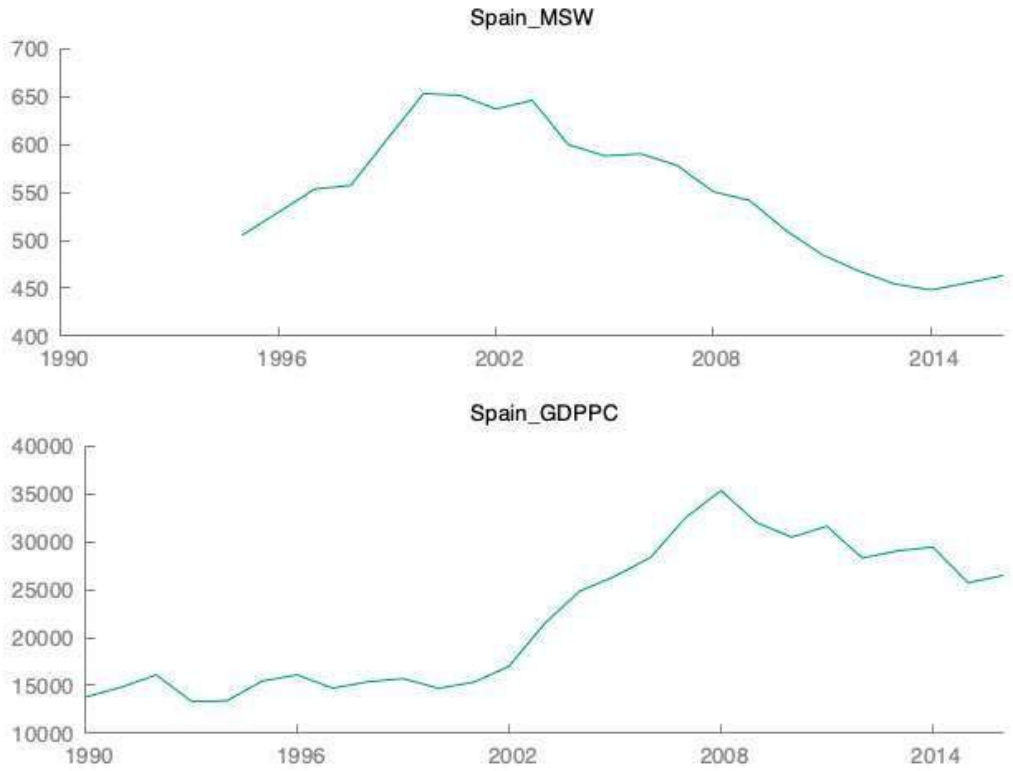
Wastewater Management in the Philippines (Corpuz, 2025)

Corpuz (2025) saw the issues in wastewater management in the Philippines. The research saw issues in waste management such as a growing population and poor treatment facilities. This results in wastewater not being treated, being discharged to water bodies. This research used policy review and environmental review as a research method. The limitation of this research is that there is a lack of technology in monitoring wastewater. This is similar because it shows the importance of water checking systems, an objective of Garbage-Guard.

Solid Waste Management and Economic Growth (Ella et al., 2022)

Figure 3

Municipal Solid Waste against Gross Domestic Product per Capita



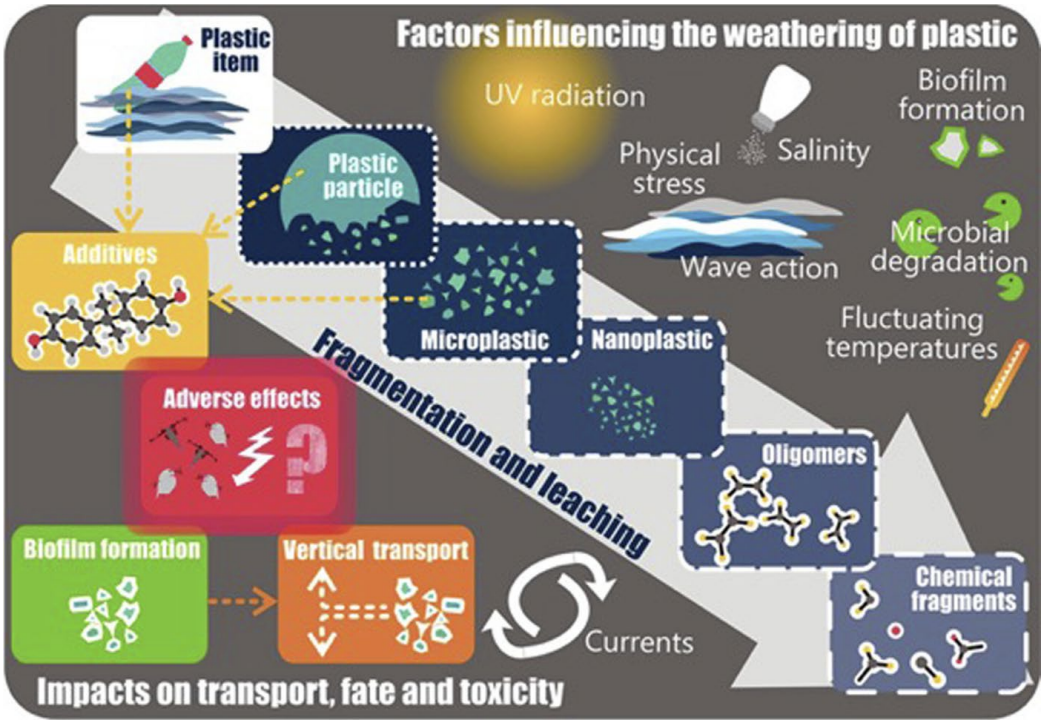
Note. From "The impact of solid waste management to the economic growth in selected OECD countries and Philippines," by D. M. Ella, D. M. Lim, B. Lu and R. Cabauatan, 2022, Journal of Economics, Finance and Accounting Studies, 4(1). CC BY.

The study by Ella et al. (2022) on the relationship between municipal solid waste (MSW) management and economic growth in selected OECD countries and the Philippines revealed that MSW has a positive effect on GDP per capita. However, the effect of recycling varies by country. The limitations of the study are its concentration on statistical data rather than real-time waste management. This study is similar because it shows the importance of managing waste.

Plastic and Microplastic Waste in Southeast Asia (Ng et al., 2023)

Figure 4

Weathering and Fragmentation of Plastic in the Marine Environment



Note. From "Plastic waste and microplastic issues in Southeast Asia," by C. H. Ng et al., 2023, Frontiers in Environmental Science, 11. CC BY.

According to Ng et al. (2023) on pollution in Southeast Asia, it is obvious that it lacks structure and poor waste collection practices are bringing an increased level of plastic pollution in water sources. The research earns a review approach and limited citations in data coverage and is dependent on secondary sources. The research is relevant in that it highlights the importance of monitoring and detection of plastic pollution, in which Garbage-Guard helps through computer vision.

B. Computer Vision Systems

Floating Object Detection on Edge Hardware (Li et al., 2024)

Figure 5

Floating Objects Detected on a Water Surface



Note. From "Detection of floating objects on water surface using YOLOv5s in an edge computing environment," by H. Li et al., 2024, Water, 16(1), 86. CC BY.

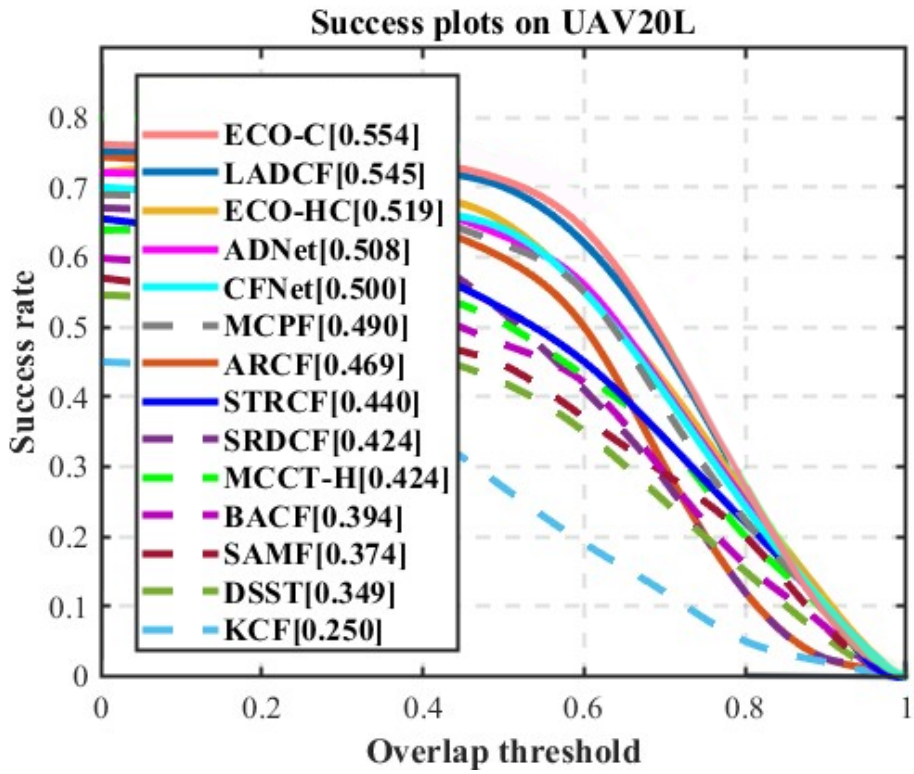
Li et al. (2024) addressed the problem this study also faces, which is running a detector on hardware small enough to sit beside the water rather than in a server room. Their improved YOLOv5s model raised detection accuracy by 15.7 per cent over the baseline, cut false detections on small targets by 83 per cent, and sustained 33 frames per second on an edge device. Two findings carry directly into Garbage-Guard. The first is that a small YOLO variant is sufficient for floating-object detection, which supports the choice

of YOLO26n over a heavier model. The second is that accuracy is lost on small and distant objects, which matches the weakness this study observes at the creek, where floating bags are the least reliably detected class.

Computer Vision Detection and Visual SLAM (Chen et al., 2024)

Figure 6

Success Rates of Visual Tracking Algorithms on a Benchmark Set



Note. From "A survey of computer vision detection, visual SLAM algorithms, and their applications in energy-efficient autonomous systems," by L. Chen et al., 2024, Energies, 17(20), 5177. CC BY.

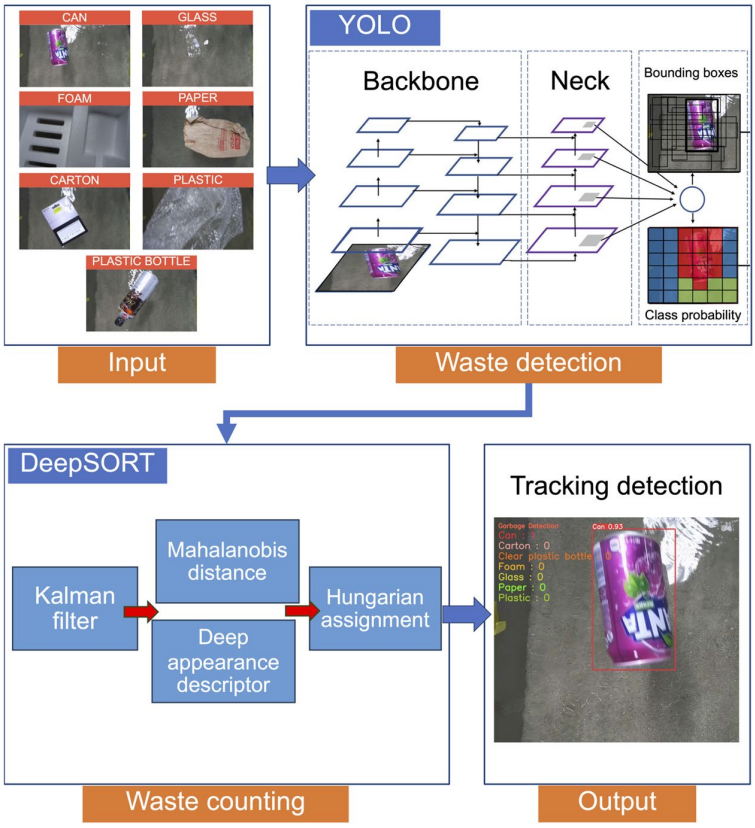
According to the study of Chen et al. (2024), computer vision serves as a foundational element in the realm of intelligent systems and AI technologies. It can effectively replace traditional sensors as the primary source of data input, acting as a pair of intelligent "eyes" that can more adeptly process data sources to drive a range of unmanned operations, such as autonomous driving, obstacle avoidance, and detection

tracking. This study is relevant to researchers' study as we integrate with computer vision to detect waste in creeks in real time.

Detecting and Tracking River-Floating Waste (Nunkhaw & Miyamoto, 2024)

Figure 7

Detection and Tracking Pipeline for River-Floating Waste



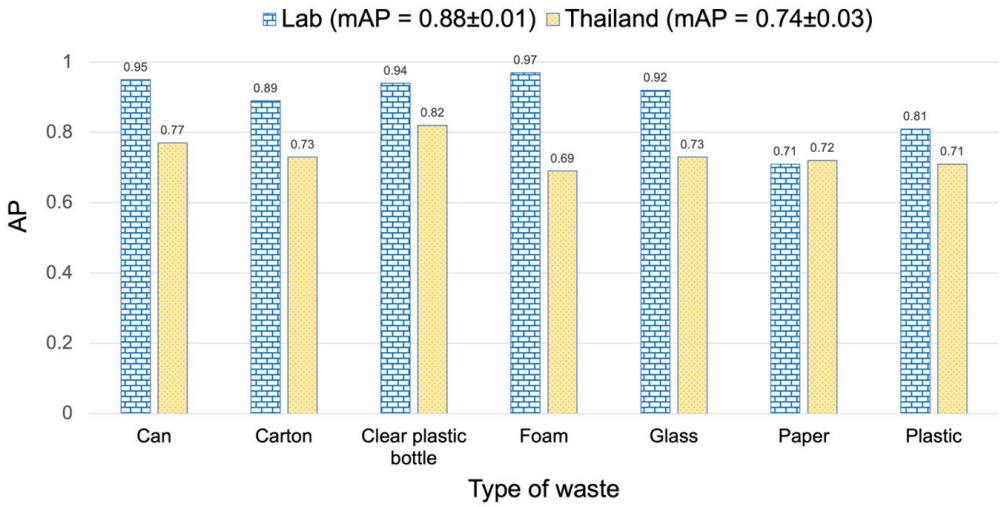
Note. From "An image analysis of river-floating waste materials by using deep learning techniques," by M. Nunkhaw and H. Miyamoto, 2024, Water, 16(10), 1373. CC BY.

Nunkhaw and Miyamoto (2024) combined a YOLOv5 detector with DeepSORT tracking to analyze river-floating waste, reporting a mean average precision at or above 88 per cent and roughly 80 per cent accuracy across three scenarios. The closest to the method used here that is published is theirs, which makes it clear that detection alone is not sufficient on moving water. If an item is moving around the frame, it should be a single identity in every subsequent frame, otherwise it is counted again if it appears in each frame.

Garbage-Guard comes to the same conclusion and to the same solution, which is to replace DeepSORT with ByteTrack. River Waste Detection in Urban Canals (Nunkhaw et al., 2025)

Figure 8

Detection Accuracy in the Laboratory Compared with the Field



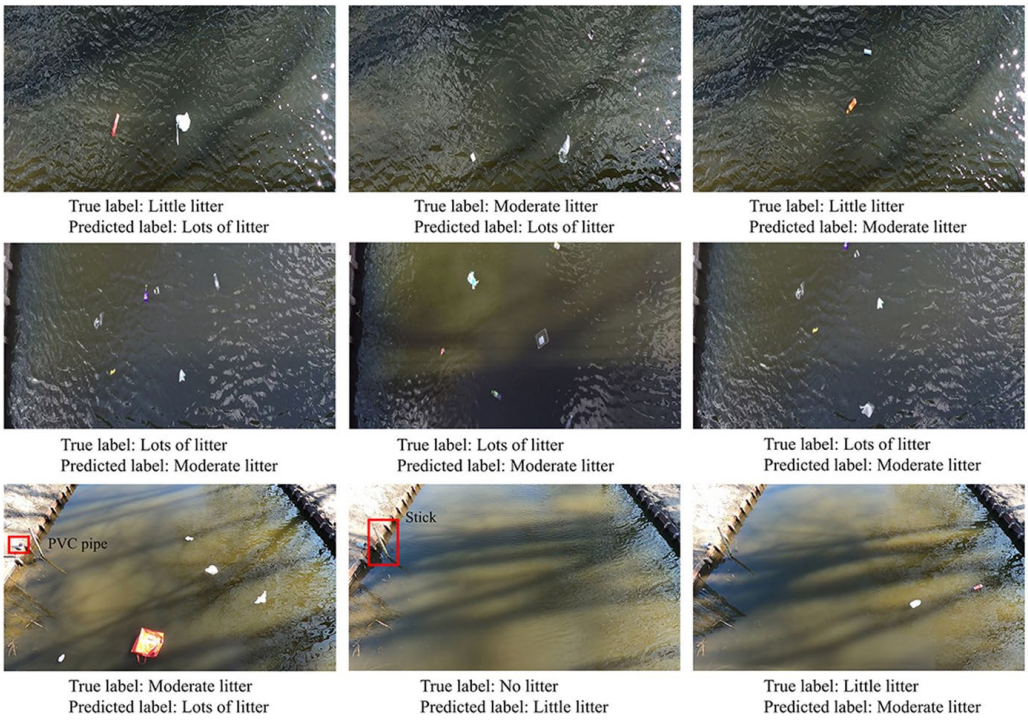
Note. From "Enhancing river waste detection with deep learning and preprocessing: A case study in the urban canals of the Chao Phraya River," by M. Nunkhaw, D. Chitwatkulsiri and H. Miyamoto, 2025, Water, 17(22), 3193. CC BY.

Nunkhaw et al. (2025) extended that work to the urban canals of the Chao Phraya River, training a YOLO detector on laboratory flume images and then adapting it to real canal conditions through a preprocessing pipeline. The study bears on Garbage-Guard in two ways. It is set in Southeast Asian urban waterways, where waste composition and water conditions resemble a Metro Manila creek more closely than European or North American river studies do. It also documents the gap between performance on controlled images and performance in the field, which is the same gap this study reports between its full test partition and its creek-only subset.

An Open Dataset for Floating Litter Detection (Jia et al., 2023)

Figure 9

Floating Litter Images with True and Predicted Labels



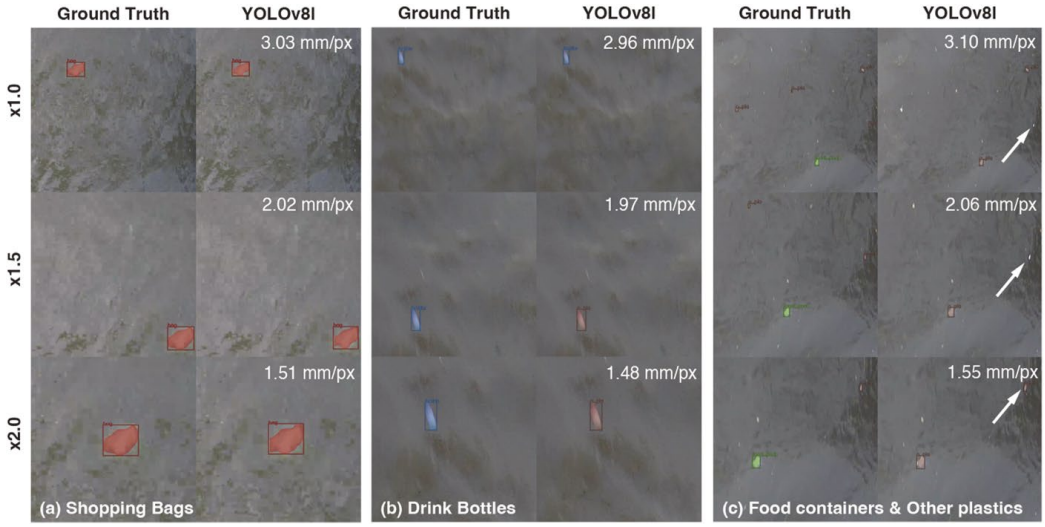
Note. From "Advancing deep learning-based detection of floating litter using a novel open dataset," by T. Jia, A. J. Vallendar, R. de Vries, Z. Kapelan and R. Taormina, 2023, Frontiers in Water, 5. CC BY.

To overcome the lack of public training data for floating litter detection, Jia et al. (2023) created an open training data set of floating litter and tested the detection performance that the data set could contribute. The issue they present is the one that drove the data collection for this study: there was not an available dataset of images that would match the type of wastes being encountered in the creeks of the Philippines. Their work presents a published explanation to why they chose to build their own dataset as opposed to taking an already existing one and adapting it.

Segmentation Compared with Detection for Floating Macroplastic (Kataoka et al., 2024)

Figure 10

Ground Truth Compared with Model Output for Three Waste Classes



Note. From "Instance segmentation models for detecting floating macroplastic debris from river surface images," by T. Kataoka, T. Yoshida and N. Yamamoto, 2024, Frontiers in Earth Science, 12. CC BY.

Kataoka et al. (2024) compared instance segmentation models for floating macroplastic in river surface images. Segmentation returns a precise outline of each object rather than a rectangle around it, which is more informative but considerably more expensive to compute. Their results support the decision taken here to use bounding-box detection instead. For counting accumulation against a threshold, what matters is the position and identity of an item rather than its exact outline, and the computation saved is what allows the model to run on a Raspberry Pi at the waterside.

Classifying Waste Fragments with Convolutional Features (Nežerka et al., 2023)

According to the study of Nežerka et al. (2023), to address improper sorting of construction and demolition waste that leads to significant environmental and economic implications, they developed a machine-learning-assisted procedure for recognizing construction and demolition waste fragments using a RGB camera. It resulted in an overall

85.9%, offering a pathway for efficient and accurate waste sorting. It can be relevant to researchers' study as it relates to detecting waste using computer vision.

Lane Detection under Varying Light (Al Noman et al., 2022)

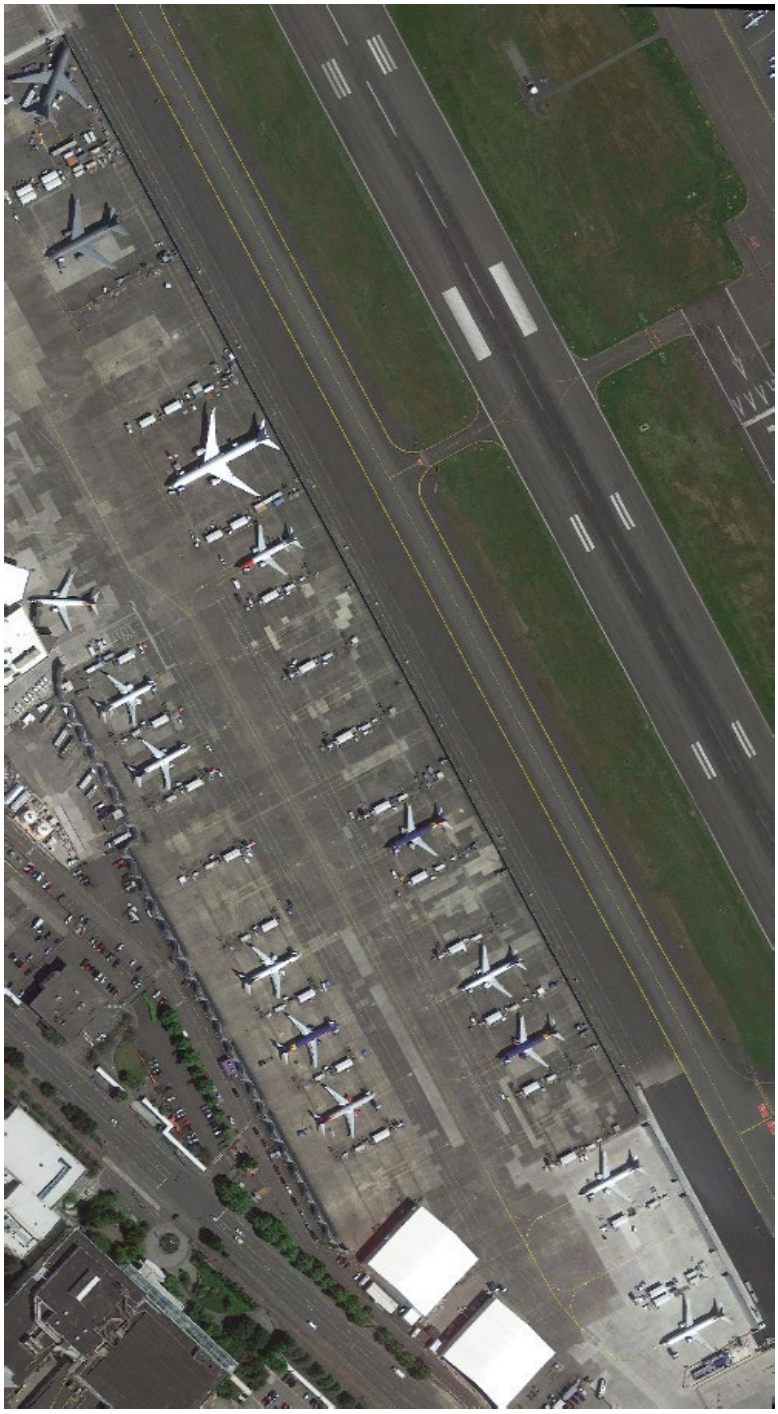
According to the study of Al Noman et al. (2022) "A computer vision-based lane detection technique using gradient threshold and hue-lightness-saturation value for an autonomous vehicle" the system uses computer vision-based technologies to identify road lines. The left and right lanes on the roads are determined via sliding window searching technique. The system works with 96% accuracy and the positive rate is 0.640%. The accuracy and the efficiency of the system could be developed by increasing computer features and applying deep learning algorithms along with the proposed study.

Object Detection in Overhead Imagery (Kang et al., 2022)

According to Kang et al. (2022), computer vision technology has enhanced the processing of objects in high-resolution overhead images taken by drones, airplanes, and satellites. The technology has been applied in real-world applications like crop analysis, economic analysis, and disaster tracking. Despite the advancements, object detection in overhead images is still a challenge due to the volume of images, resolution, size of objects, and complexity of the background.

Figure 11

Overhead Imagery of the Kind Used for Object Detection



Note. From "A survey of deep learning-based object detection methods and datasets for overhead imagery," by J. Kang, S. Tariq, H. Oh and S. S. Woo, 2022, IEEE Access, 10. CC BY.

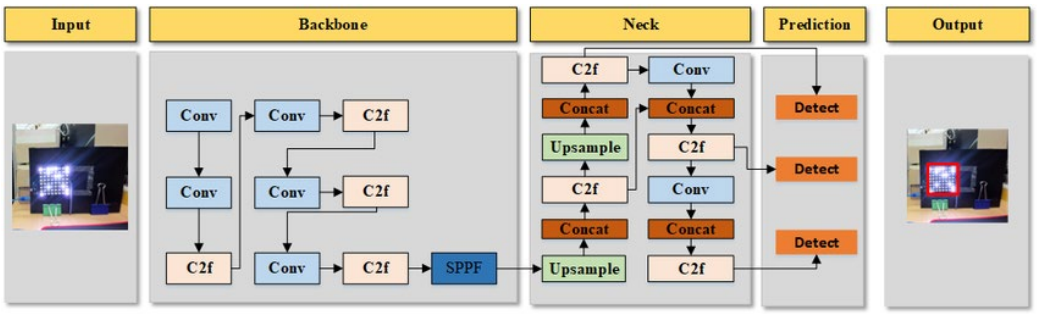
A Review of YOLO Versions for Object Detection (Murat & Kiran, 2025)

Similarly, Murat & Kiran (2025) described that object detection algorithms have undergone a tremendous transformation with the application of deep learning technology. Among the most popular algorithms is YOLO, which is famous for its fast and accurate real-time object detection. The researcher has reviewed the various versions of YOLO, including YOLOv1 to YOLOv11, and described the advancements in the architecture and performance of the algorithm.

The Internal Architecture of YOLOv8 (Yaseen, 2024)

Figure 12

Architecture of the YOLOv8 Detector



Note. From "What is YOLOv8: An in-depth exploration of the internal features of the next-generation object detector," by M. Yaseen, 2024, arXiv. CC BY.

Moreover, Yaseen (2024) has examined YOLOv8 and described the advancements of the algorithm over the previous versions. The algorithm applies advanced architectures such as CSPNet and FPN+PAN, along with an anchor-free technique, to enhance the feature extraction and multi-scale detection. The algorithm has been evaluated using various datasets, including Microsoft COCO, which shows the accuracy and real-time detection capabilities of YOLOv8.

C. Smart Environmental Monitoring Systems

Real-Time Environmental Monitoring with IoT Sensors (Narayana et al., 2024)

In the research conducted by Narayana et al. (2024), environmental conditions are made up of both living and non-living components that affect human life. The non-living components include land, water, stones, atmosphere etc., and living components include organisms like plants, woodlands, fisheries and wildlife etc. Environmental parameters, like temperature, humidity, pressure, air quality, greenhouse gases, wind speed and direction, rainfall, soil pH, soil moisture, water availability, light, turbidity, etc., influence people's lives, especially in the case of workers who practice activities such as agriculture, navigation, traveling, and mining.

In addition to that, to address the issue of monitoring environmental variables and to mitigae the challenges that come with monitoring environmental parameters, IoT technology was used to develop IoT-based smart environmental monitoring systems. This technology uses microcontrollers and other technologies to monitor environmental parameters from different locations and send the data to a website for monitoring and observation remotely using IoT technology. IoT technology is a dynamic system that observes and responds to environmental variables and conditions (Narayana et al., 2024)

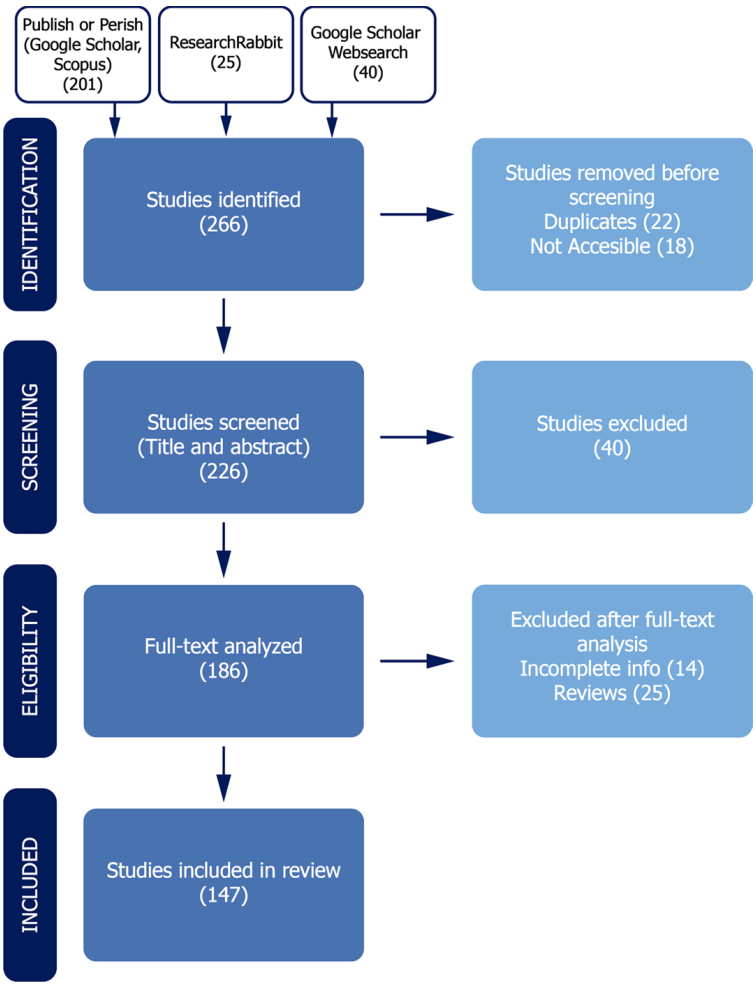
Sensors in Smart City Systems (Zeng et al., 2024)

In the study of Zeng et al. (2024), the integration of IoT technology into smart city systems has several advantages in the management of the environment, as it allows real-time monitoring of important parameters such as air quality, water quality, noise levels, soil quality, and others. This has enabled city administrators to proactively manage the resources of the city in a cost-effective manner. These parameters have collectively proven the importance of IoT sensors in achieving sustainability in the environment.

IoT and Artificial Intelligence in Air Quality Monitoring (Garcia et al., 2025)

Figure 13

Selection of Studies for a Systematic Review of Air Quality Monitoring



Note. From "Advancements in air quality monitoring: A systematic review of IoT-based air quality monitoring and AI technologies," by A. Garcia, Y. Saez, I. Harris, X. Huang and E. Collado, 2025, Artificial Intelligence Review, 58. CC BY.

Expanding on these capabilities, Garcia et al. (2025) show how Internet of Things (IoT) technologies have greatly improved environmental monitoring by allowing a network of low-cost sensors to continually produce large amounts of real-time data on air quality. This is further enriched by using these IoT systems with the Artificial Intelligence techniques like machine learning and deep learning. This assists to ensure as accurate an

environmental assessment as possible. IoT-based monitoring system can monitor various air quality parameters like PM, CO_2, NO_2, ozone, VOCs etc. This makes them an efficient means of conducting a thorough surveillance of the environment. The integrated IoT/artificial intelligence-based systems are evident examples of how these systems can enhance the decisions made with respect to environmental management.

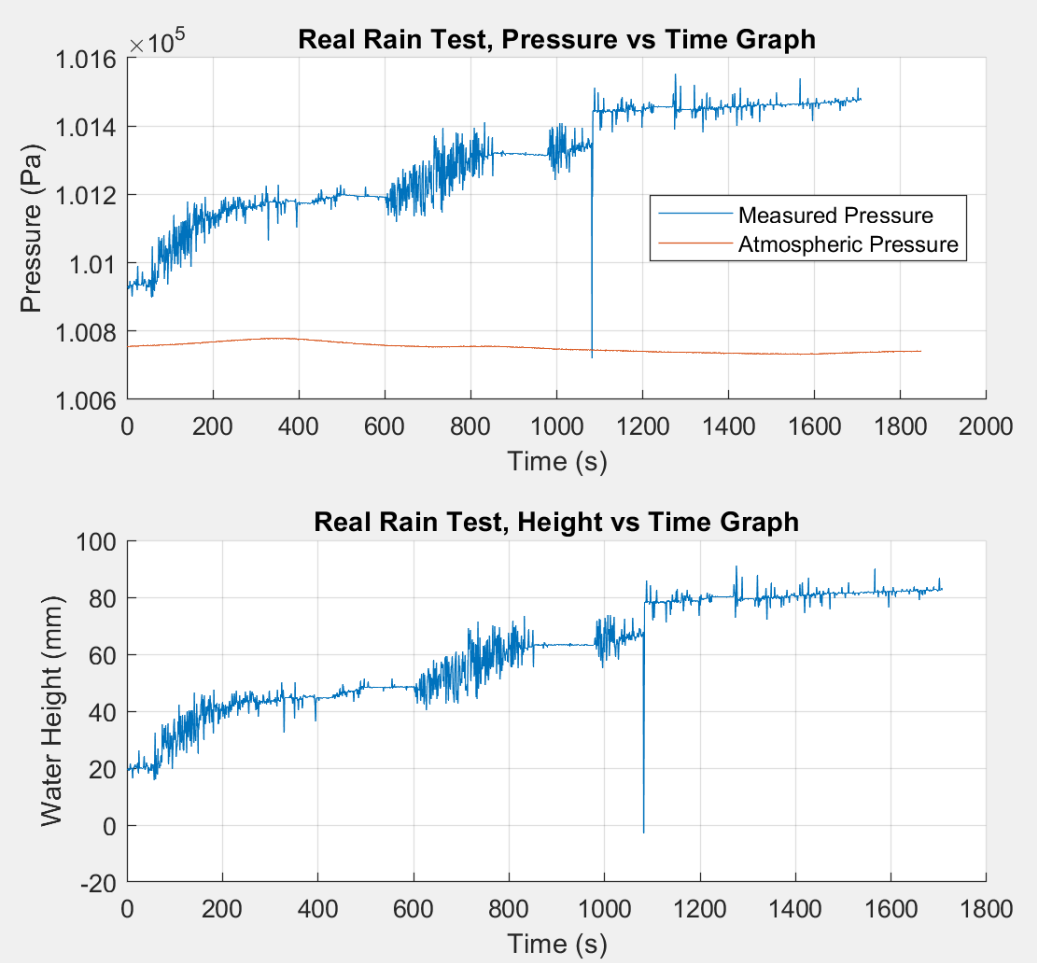
Machine Learning and IoT for Pollution Prediction (Gryech et al., 2024)

In the recent literature of Gryech et al. (2024), it has been observed that there is a rising trend towards the inclusion of IoT technologies with machine learning methods for better outdoor air pollution prediction and forecasting. Though the traditional air pollution stations are accurate, the high cost of these stations has encouraged the development of IoT-based low-cost sensors, along with machine learning methods that are capable of predicting the levels of outdoor air pollution through time-series, feature-based, and spatio-temporal methods, although the limitations of the lack of diversity in the data and the context-specific features have to be addressed, as identified by the recent literature.

Urban Flood Monitoring with LoRaWAN Sensors (Te et al., 2024)

Figure 14

Water Level and Pressure Recorded during a Rainfall Event



Note. From "A smart IoT urban flood monitoring system using a high-performance pressure sensor with LoRaWAN," by M. C. L. Te et al., 2024, HighTech and Innovation Journal, 5(4). CC BY.

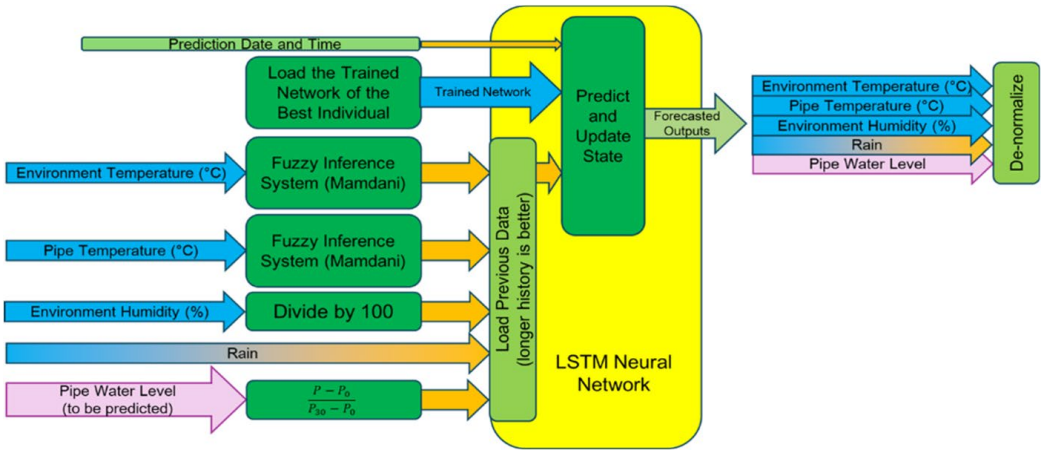
One of the significant environmental issues experienced in the Philippines has been flooding, which has been attributed to monsoon rains, the topography of the area, and inadequate drainage systems. This has been a major cause of loss of lives and properties, especially in the urban area. In order to address the issue of delay associated with flood monitoring, Te et al. (2024) proposed the development of a smart IoT-based flood monitoring system for an urban area. This would entail the incorporation of efficient

pressure sensors, LoRaWAN, and solar power. From the reviewed background of the research, it is clear that there is a need to utilize IoT to address the issue of disaster, as it would allow for efficient monitoring to enhance the flood response strategy (Te et al., 2024).

Secured Flood Monitoring and Forecasting (Arante et al., 2025)

Figure 15

Neuro-Fuzzy Network for Flood Forecasting



Note. From "Development of a secured IoT-based flood monitoring and forecasting system using genetic-algorithm-based neuro-fuzzy network," by H. R. C. Arante et al., 2025, Sensors, 25(13), 3885. CC BY.

The monitoring and forecasting of floods in the Philippines are important due to the occurrence of heavy rains and typhoon scenarios, which cause the rapid rise of water levels. In this regard, Arante et al. (2025) contextualized their research by highlighting the limitations of the existing flood monitoring systems, which are not effective in the Philippines, and the need to develop a secure IoT infrastructure to transmit data on water levels to aid in the prediction of floods. The authors proposed a neuro-fuzzy LSTM machine learning model to improve the accuracy of flood event prediction, which is a form of smart environmental monitoring system in a water-related disaster scenario.

2.3 Review of Related Patents

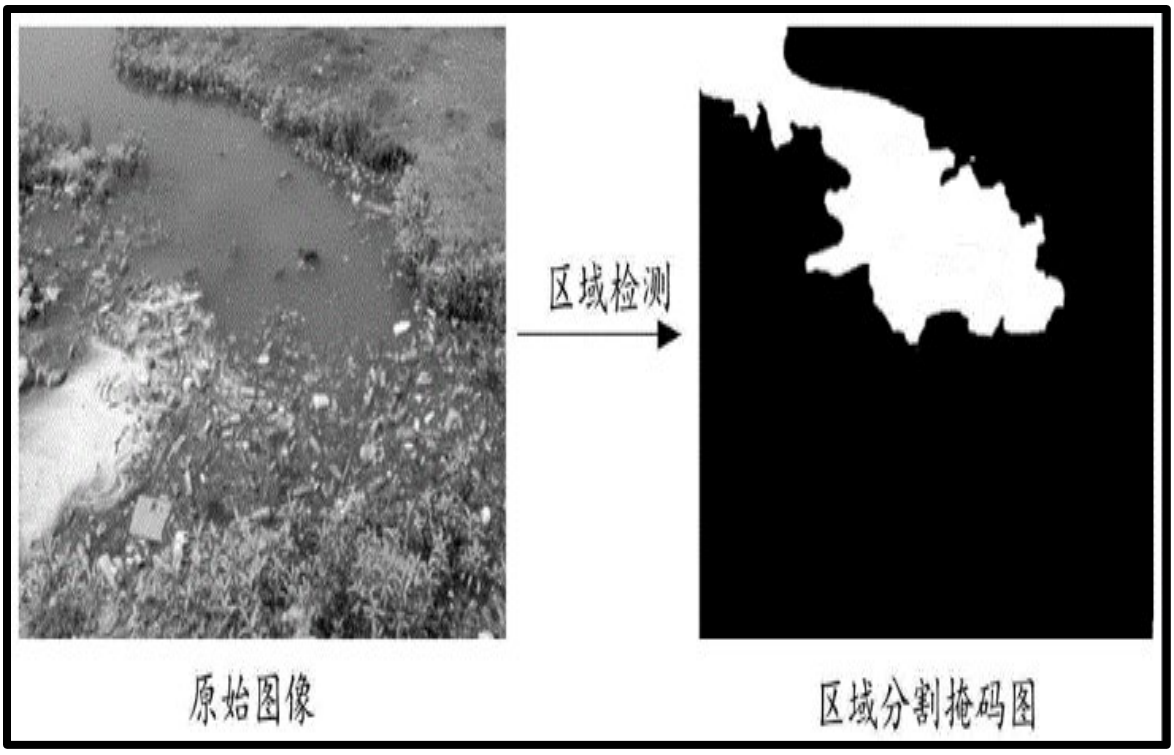
Table 1

Table of Related Patents

Patent Title	Author	Adopted Feature	Enhanced Feature
Garbage Detection Method, Related Equipment and Device	Li & Wang (2020)	Image recognition to detect garbage	Enables automatic garbage detection in areas.
Vision garbage classification system based on TensorFlow	Wu et al. (2019)	The AI system classifies garbage usingTensorFlow.	Using similar A system, it improves garbage recognition using deep learning model like YOLOv8
Smart City Waste Collection and Management using IoT	Chordiya et al. (2020)	IoT Sensors monitor garbage compiling systems.	Using a camera for real-time monitoring of garbage
Cleaning Unit Using Deep Learning for Floating Trash Detection	*Hyun-dong et al.* (2019)	Uses object detection to identify trash	Enhances garbage identification using AI algorithms.
Drainage monitoring apparatus and system	Campbell et al. (2022)	Water level sensors monitor drainage conditions and detect faults.	Detects garbage as a cause of drain blockage in waterways.
System for monitoring water drainage	Den Hartog & Voncken (2017)	Sensors monitor water levels in drain systems.	Detects water blockage and water conditions.
Detection and monitoring of waste disposal using artificial intelligence	Liu et al. (2022)	AI monitors garbage in the environment	Tracks garbage data using automated analysis.
Flood warning system	Torres et al. (2024)	Sensors detect rising water levels.	Alerts user when garbage buildup cause flooding
Method and System for Flood Early Warning in Smart Cities Based on the Internet of Things	Shao et al. (2023)	IoT sensors monitor rainfall and water levels	Garbage detection as an early indicator of flooding risk
Apparatus and Method for Waste Monitoring and Analysis	Balachandran & Talavera (2023)	Sensors and cameras monitor garbage levels in containers.	Automates garbage detection using Computer Vision analysis

Figure 16

Garbage Detection Method, Related Equipment and Device

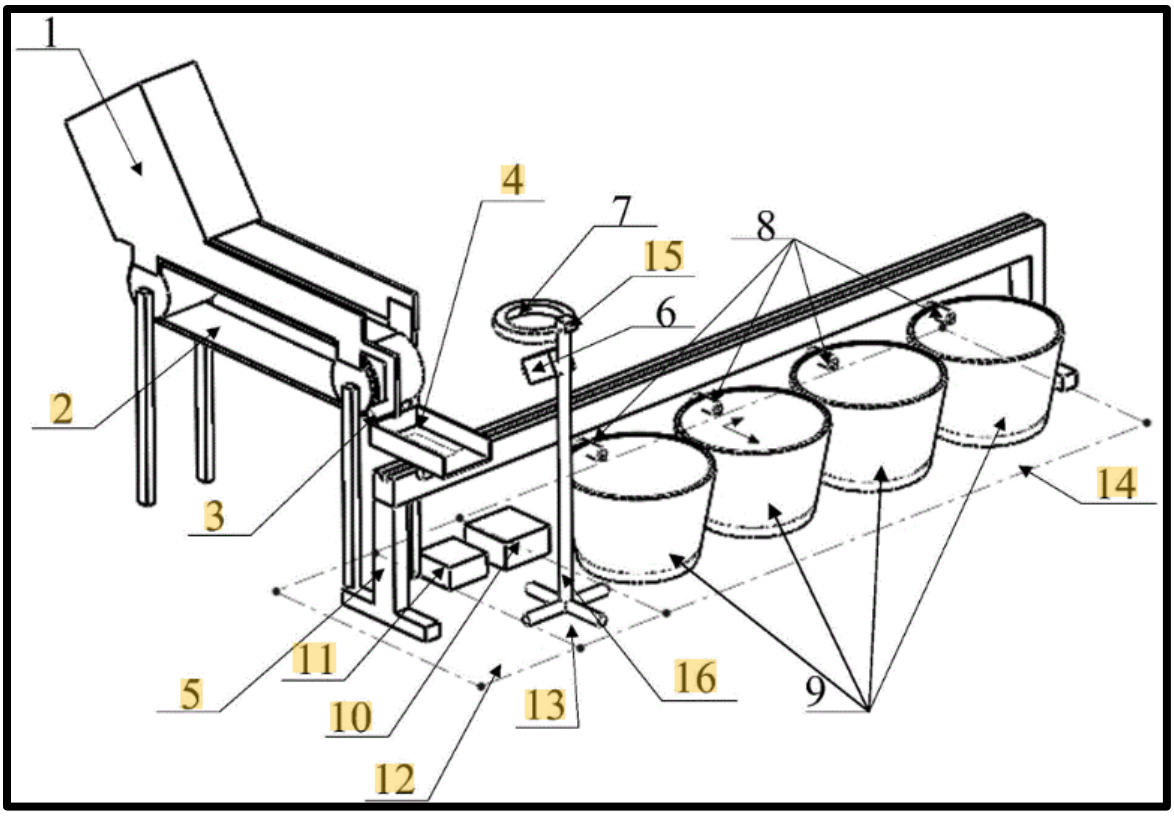


In Figure 16, Chinese researchers have made a garbage detection method using image recognition. The system takes pictures of monitored rivers and processes them to find garbage.

This technology can be used in different urban areas such as lakes and parks. It analyzes captured images to think whether garbage objects are visible in the environment. In this patent's case, it relates to Garbage-Guard because it can use computer vision to detect garbage, similar to the researcher's proposed system that detects garbage in creeks.

Figure 17

Vision garbage classification system and classification method based on TensorFlow



In Figure 17, this patent which uses computer vision, is a Vision Garbage Classification System and Method based on TensorFlow made by Wu et al. (2019). The system uses images of garbage items and puts them into a model trained on garbage images. Methods include data collecting, image processing, and automatic classification of garbage categories.

This patent is similar to the researcher's present study because it also uses the application of computer vision and machine learning techniques but for automated garbage detection.

Figure 18

WCM-Smart City: WASTE COLLECTION AND MANAGEMENT UNDER SMART CITY USING IOT

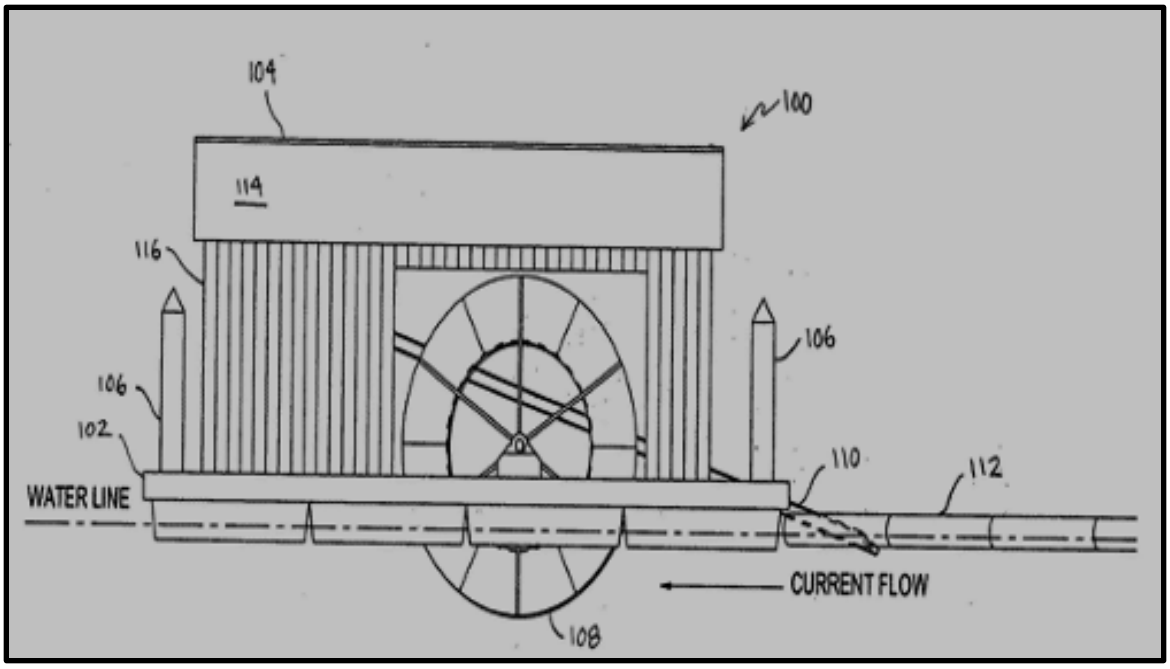


Figure 18 is a patent from Chordiya (2020), describing their study as a waste collection system made for grabbing garbage in rivers and streams. This invention uses a conveyor mechanism that collects waste and transports it into a container. It incorporates monitoring systems with sensors that detect waste levels and can transmit that information to a platform.

This patent demonstrates the use of automatic methods for technologies in managing garbage, it is similar to the researcher's Garbage-Guard system that aims to find garbage in creeks using computer vision technology.

Figure 19

Cleaning Unit Using Deep Learning for Floating Trash Detection

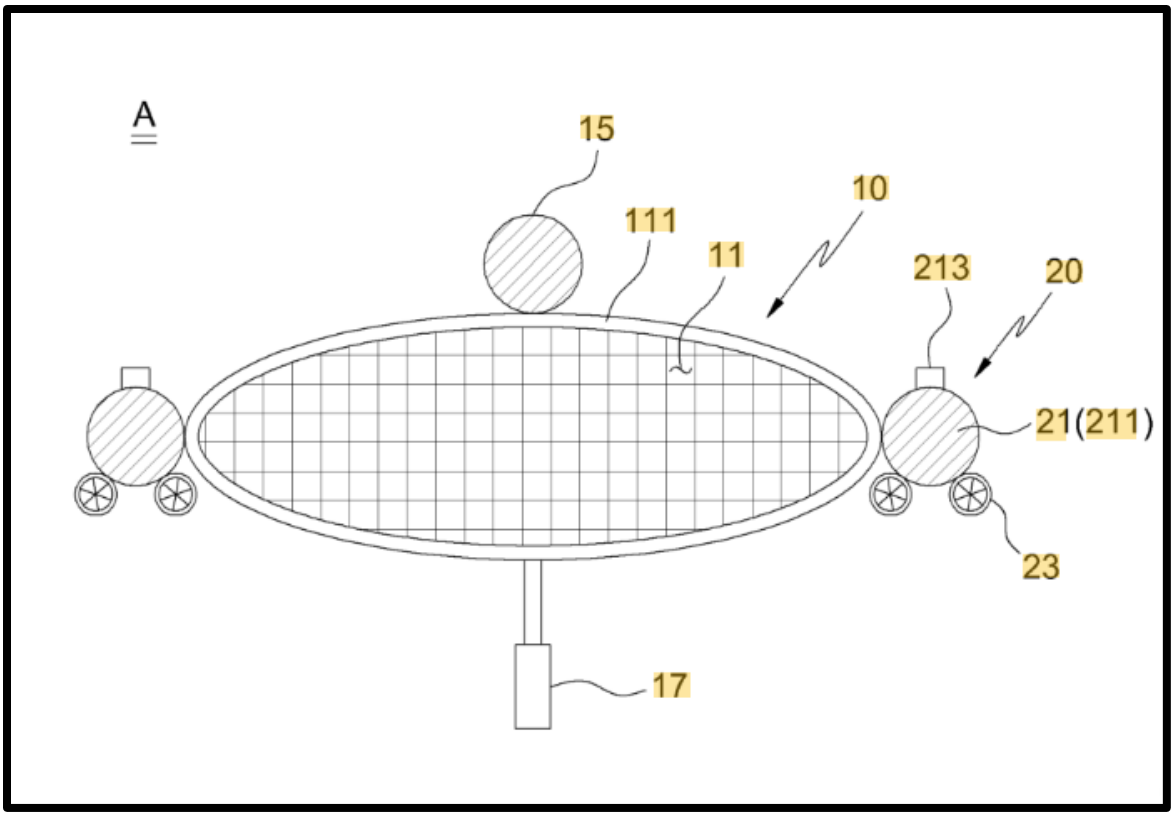


Figure 19 is a similar patent that is related to garbage detection from South Korea. It is a water-surface cleaning unit that uses deep learning technology to detect floating garbage made by Kim et al. (2019). The system integrates a camera module with a deep-learning model which helps in locating garbage on water surfaces. Then, it activates a mechanism that collects the debris from the water.

This type of invention shows how relevant it is to the researchers' study because it is similar to how artificial intelligence is used for garbage detection in water environments.

Figure 20

System for Monitoring Water Drainage

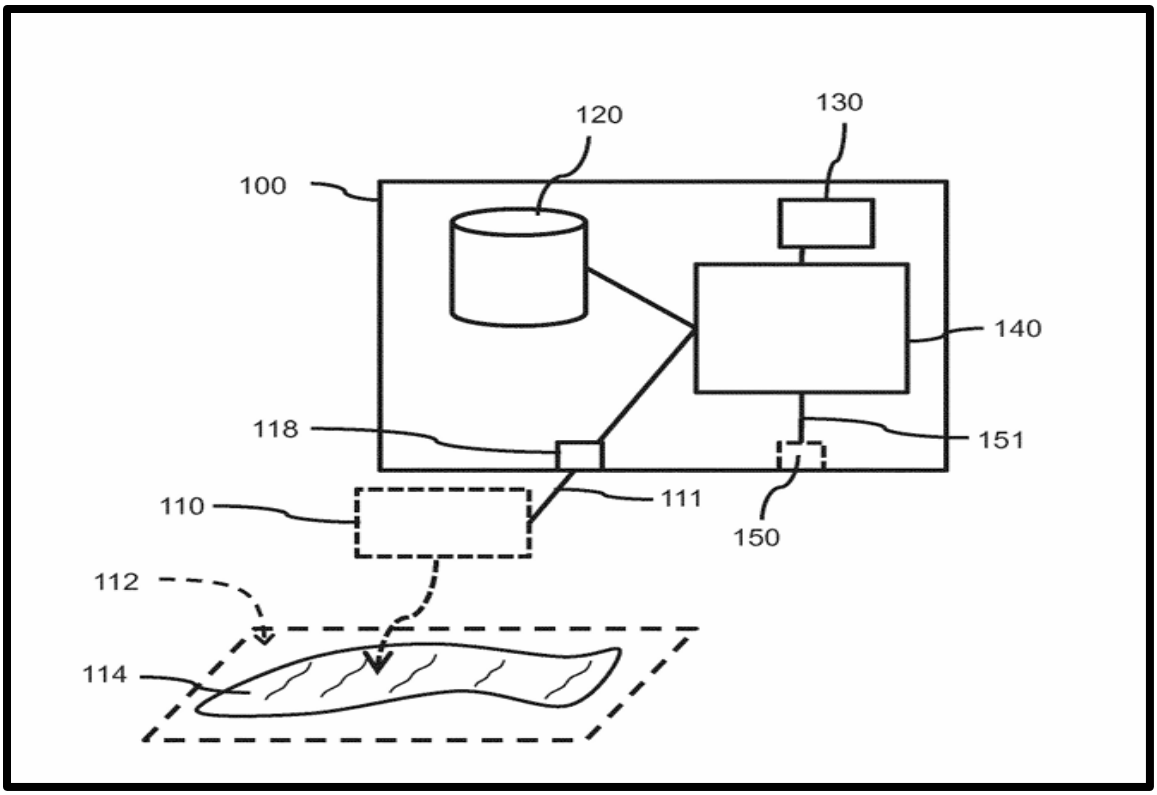


Figure 20, in terms of smart drainage systems, this figure from a patent made by Den Hartog and Voncken (2017) patented a system that is designed to check water drainage environmental sensors. The system collects data from water presence on surfaces and compares it with predicted water drainage conditions based on what happens during rainfall. If the detected water remains longer than usual, the system makes warning signals about potential drainage problems.

This type of technology integrates sensors, weather information, and data processing to check drainage conditions. This type of patent is related to Garbage-Guard because both of the systems required to monitor water environments, by detecting garbage being collected in creeks, which may cause drainage issues.

Figure 21

Drainage Monitoring Apparatus and System

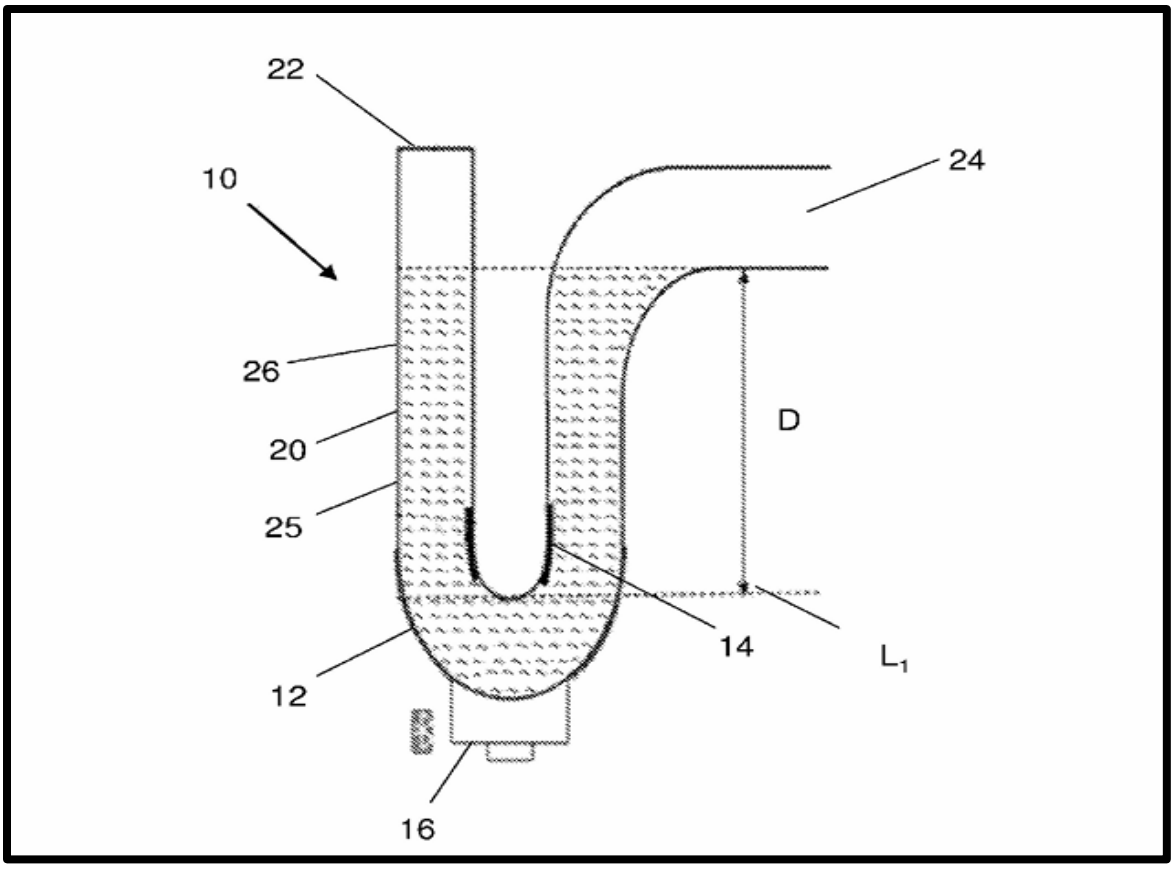


Figure 21 is another patent related to smart drainage monitoring systems, is this patent from Campbell et al. (2022), which showcases a drain monitor apparatus designed to look at water levels within building drainage systems. The invention uses a water trap equipped with a water level sensor that checks the level of water inside the drainage trap. When the water level drops to a specific point, the sensor sends a signal to a system or mobile application that can make audible alerts to inform that there are drainage faults.

This technology is relevant to Garbage-Guard since it is similar to how the systems monitor drainage systems. However, instead of detecting water levels, it detects the various amounts of garbage in creeks using computer vision.

Figure 22

Detection and Monitoring of Waste Disposal Using Artificial Intelligence

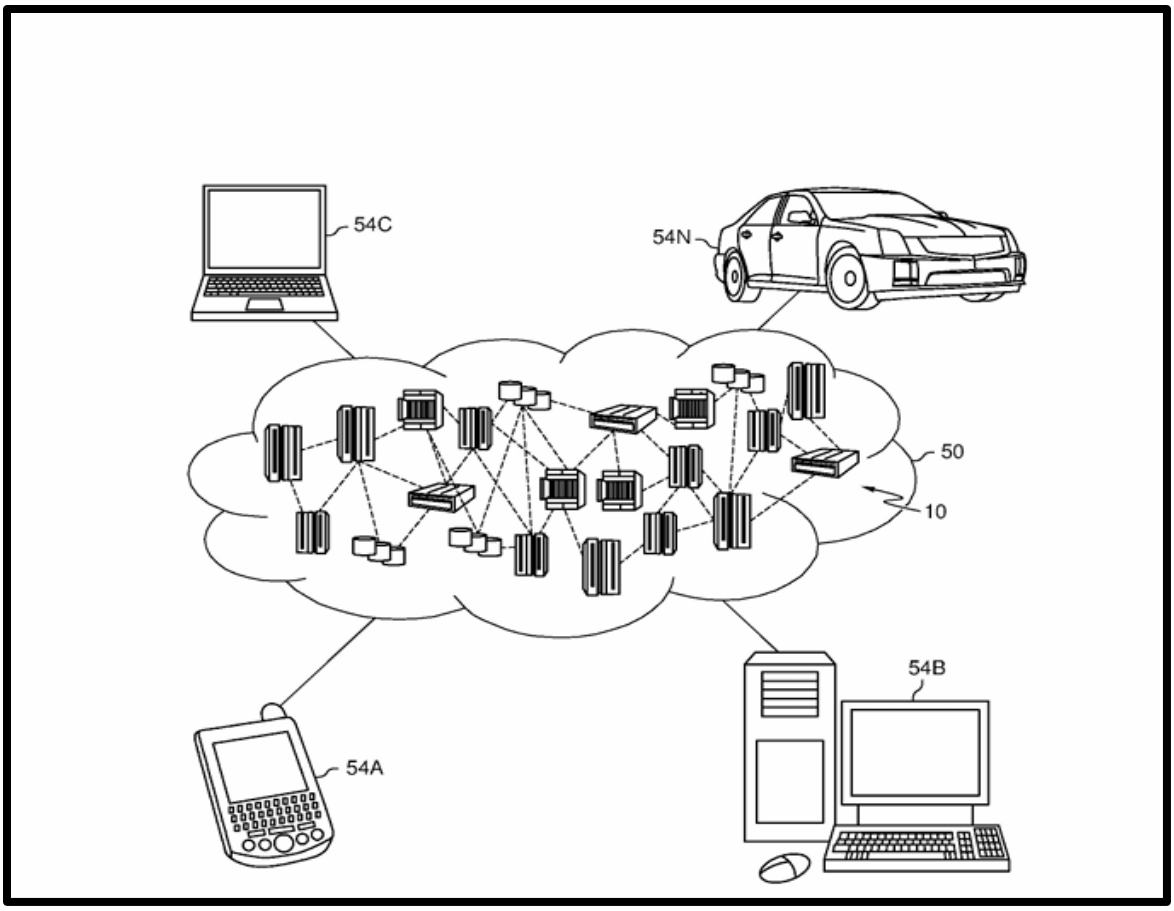


Figure 22 patent by Liu et al. (2022) made an artificial intelligence system for detecting waste in environmental locations, just like natural environments. The system collects images using sensors or cameras and gives them to machine-learning algorithms that check waste objects. Then, the system tracks waste accumulation patterns and finds the location and size of garbage in monitored areas.

This patent is similar to Garbage-Guard's usage of Artificial Intelligence and image analysis to find garbage in different locations.

Figure 23

Flood Warning System

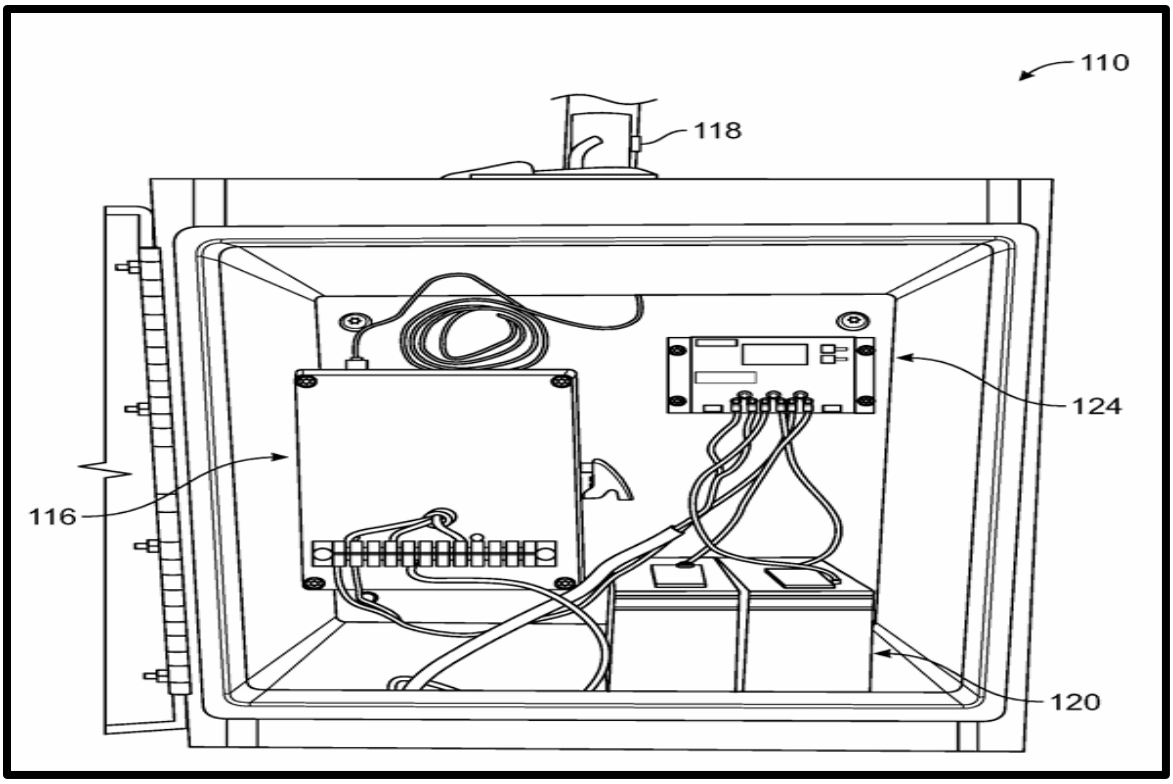
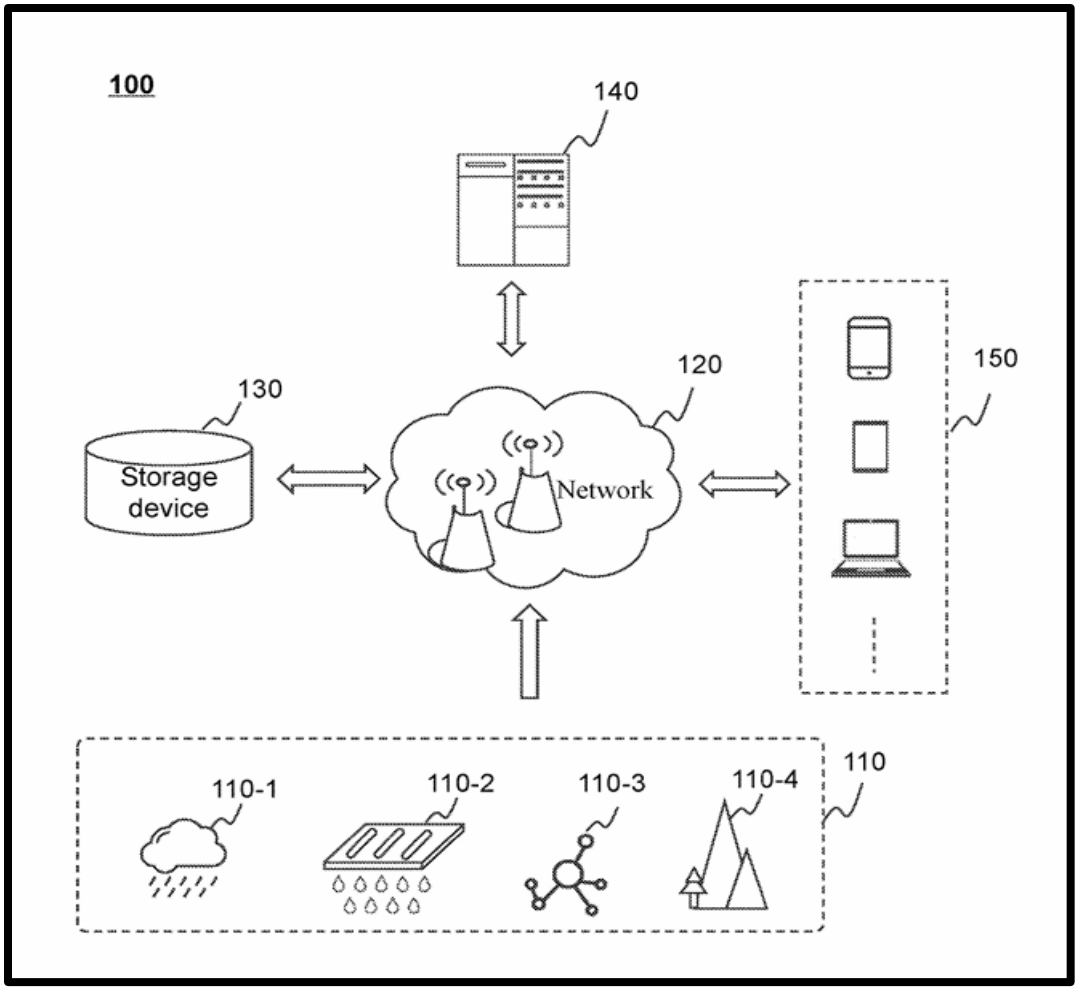


Figure 23 patent by Torres et al. (2024) made a flood warning system to provide early warnings for floods using monitoring systems. It includes multiple of these in monitoring stations near rivers or drainage channels. Each of these are equipped with sensors that monitor water level and rainfall accumulation. That collected data is connected wirelessly to a flood warning station, where it is checked to determine whether the water level exceeds the thresholds. When a flood risk has been shown, the system will make visual alerts that are sent to users through mobile devices.

This patent is useful and similar to Garbage-Guard's system, since warning systems are important in monitoring the environment and can notify when the creeks are filled with garbage.

Figure 24

Method and System for Flood Early Warning in Smart Cities Based on the Internet of Things

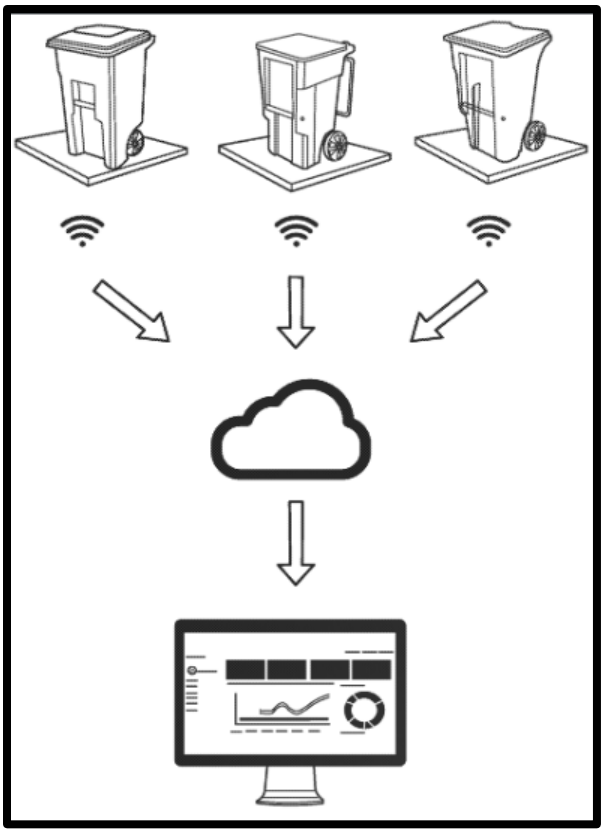


The patent at Figure 24 from Shao et al. (2023), made a smart flood early warning system that has Internet of Things (IoT) with data to detect potential flooding conditions in urban places. This system collects precipitation information from weather stations and combines it with the data from the sensors to estimate water level in specific areas.

This invention is similar to Garbage-Guard where they involve environmental monitoring and alerts.

Figure 25

Apparatus and method for waste monitoring and analysis



In Figure 25, a patent by Balachandran & Talavera (2023) invented a waste monitoring and analysis system, made for analyzing waste collected in containers. It uses sensors such as load cells and cameras to measure the weight and type of waste material placed inside bins. That data is collected and given to a cloud platform where it can be analyzed. With this information, it can define the pattern of the trash disposals, and make reports about waste generation.

This patent is relevant to Garbage-Guard because it shows how sensor-based systems can monitor and analyze data in real time. The invention focuses on detecting waste garbage collection in creeks using computer vision technology.

2.4 Synthesis of the Review

The literature reviewed in this chapter converges on four points. First, the volume of plastic that enters the ocean through rivers is also very large, a very few rivers account for the majority of the plastic and the highest concentration of plastic is found at the lowest collection sites. Meijer et al. (2021) state that more than 1000 rivers are responsible for 80% of riverine plastic emission, with Southeast Asia (Ng et al., 2023) and the Philippines (Corpuz, 2025) being those where collection infrastructure is not at par. The problem this study addresses is therefore a documented and measured one rather than a local inconvenience.

Second, it's not scalable to manual inspection. Studies, which have attempted to quantify waste in waterways, are dependent on a survey campaign, a satellite pass or a modelled estimate, all of which fail to return a reading for the operator to take action on the same afternoon.

Thirdly, convolutional object detection has reached a certain level of maturity and can handle the task. Murat and Kiran (2025) and Yaseen (2024) report the accuracy and speed improvements of all the YOLO versions over the past 10 years, while Nežerka et al. (2023) demonstrate that convolutional classification is superior to hand-selected feature extraction with waste fragments in particular. Detection of floating waste is not an open research question.

Fourth, the supporting pieces are set up separately. Li et al. (2024) show a ‘Very small YOLO’ (VsYOLO) that runs on an edge device to identify floating objects at a frame rate of 33 fps. Nunkhaw and Miyamoto (2024) combine detection with tracking to ensure that an item passing through the frame is only counted once. Nunkhaw et al. (2025) continue this practice to urban canals in Southeast Asia and report the difference between lab and field. Jia et al. (2023) tackle the lack of public floating-litter data, while Kataoka et al. (2024) compare segmentation with bounding-box detection. Narayana et al. (2024),

Zeng et al. (2024) and Te et al. (2024) prove that environmental sensing on a limited hardware platform coupled with real-time alerting is commonplace.

None of the studies examined here draw the components together in the way needed for this project. The waste-detection work classifies images without any time series accumulation. Accumulation work is the work that accumulates items in containers instead of on moving water. The environmental monitoring work alerts based on scalar sensor readings and not on the count of them. Not all of the systems reviewed have an operator trained detector on the edge at creek level, limit count to an operator defined area of water, maintain one identity per item across frames, and trigger a threshold alert on that result. Table 2 presents that gap in reference to what this study posits.

Table 2

Proposed Solution Matrix

Limitation in the examined literature	Representative source	How Garbage-Guard responds
No detection of waste over the time period, only at classification.	Nežerka et al. (2023); Magyar et al. (2023)	The objects in the monitored area are counted, smoothed across 15 frames and compared to an operator-set threshold; thus, the output is a sustained accumulation reading and not a label per frame.
Waterway monitoring requires manual survey or satellite passes, which results in the readings being periodic.	Meijer et al. (2021); Victoriano et al. (2022)	Continuous processing and monitoring of a fixed camera feed, and logging of every event with the time recorded on-site
Counting systems assume a contained, static scene	Balachandran and Talavera (2023)	Region filtering helps to limit the counting within the traced region of water, and tracking helps to maintain a one-to-one correspondence between one floating object and one identity.
Alerting in the environment is based on scalar thresholds on sensors and not visual observation and evidence	Te et al. (2024); Arante et al. (2025)	The alert is based on a visual count and includes a saved snapshot and the reading that triggered the alert can be viewed.
Detection models are published on curated datasets and not deployment conditions	Li et al. (2024); Nunkhaw et al. (2025)	The model is reported on a frozen held-out partition, and a creek-only subset, so field performance is reported separately.

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